XXXXX

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only

AKuriT

(RIFAMPIN AND ISONIAZID TABLETS)

DESCRIPTION

AkuriT is brick red coloured, capsule shaped, biconvex film coated tablets, break line on one side and plain on the other side. Akurit is a fixed dose combination of Rifampin and Isoniazid, for use in the treatment of tuberculosis. This has been developed for the patient compliance.

COMPOSITION

Each film-coated tablet contains
Rifampin USP………..150 mg
Isoniazid USP…………..75 mg

CLINICAL PHARMACOLOGY

Rifampin inhibits DNA-dependent RNA polymerase activity in susceptible cells. Specifically, it interacts with bacterial RNA polymerase but does not inhibit the mammalian enzyme. This is the mechanism of action by which rifampin exerts its therapeutic effect. Rifampin cross-resistance has only been shown with other rifamycins.

In normal subjects the T ½ (biological half-life) of rifampin in blood is approximately 3 hours. Elimination occurs mainly through the bile and, to a much lesser extent, the urine.

Isoniazid

Isoniazid acts against actively growing tubercle bacilli. After oral administration isoniazid produces peak blood levels within 1 to 2 hours, which decline to 50%, or less within 6 hours. It diffuses readily into all body fluids (cerebrospinal, pleural, and ascitic fluids), tissues, organs, and excreta (saliva, sputum, and feces). The drug also passes through the placental barrier and into milk in concentrations comparable to those in the plasma. From 50 to 70% of a dose of isoniazid is excreted in the urine in 24 hours.

Isoniazid is metabolized primarily by acetylation and dehydrazination. The rate of acetylation is genetically determined. Approximately 50% of Blacks and Caucasians are "slow inactivators"; the majority of Eskimos and Orientals are "rapid inactivators."

The rate of acetylation does not significantly alter the effectiveness of isoniazid. However, slow acetylation may lead to higher blood levels of the drug, and thus an increase in toxic reactions.

Pyridoxine deficiency (B_6) is sometimes observed in adults with high doses of isoniazid and is considered probably due to its competition with pyridoxal phosphate for the enzyme apotryptophanase.

Pharmacodynamic

Pharmacotherapeutic group: Antimycobacterials, combinations of drugs for treatment of tuberculosis ATC Code: J04AM02.
Mechanism of action
In vitro, rifampicin is bactericidal against a wide range of organisms, including Mycobacterium tuberculosis. The mode of action is by inhibition of DNA-dependent RNA polymerase, inhibiting transcription. In tuberculosis, rifampicin is bactericidal for both intracellular and extracellular microorganisms. Microbial resistance may occur, and is a result of alterations in the target enzyme (RNA polymerase). Isoniazid is highly active against Mycobacterium tuberculosis. It is bactericidal in vitro and in vivo against actively dividing tubercle bacilli. Its primary action is to inhibit the synthesis of long chain mycolic acids, which are unique constituents of the mycobacterial cell wall. Resistance to isoniazid occurs rapidly if it is used alone in the treatment of clinical disease due to mycobacteria.

Pharmacokinetic properties

The absorption characteristics of Akurit have been determined after administration of tablets of Akurit in healthy volunteers in the fasting state as follows:

Pharmacokinetic variable	Mean value ± standard deviation (*)	
	Rifampicin	Isoniazid
Maximum concentration (C_{max}) ng/ml	12232 ±3129 (11866)	4649 ± 1713 (4313)
Area under the curve ($AUC0_{0-\infty}$), a measure of the extent of absorption ng.hour/ml	91952 ± 22974 (89462)	15719 ± 7973 (13539)
Time to attain maximum concentration (t_{max}) hour	3.60 ± 0.93	0.68 ± 0.55

*geometric mean

Pharmacokinetics of Rifampicin and Isoniazid

	Rifampicin	Isoniazid
Absorption		
Absolute bioavailability	90 - 95%	NA*
Oral bioavailability	> 90%	> 80%
Food effect	No effect on extent of absorption. Rate of absorption is reduced.	Reduced.
Distribution		
Volume of distribution (mean)	55 L	43 L
Plasma protein binding *in vitro*	60- 90%	< 10%
Tissue distribution	CSF concentrations are in the same order of magnitude as the unbound concentrations in plasma. Concentrations in liver, spleen, kidneys and lung tissue are higher than serum concentrations. Penetrates into vaginal and cervical tissue and into cervicovaginal fluid. Passes the placenta; serum concentration in fetes are about 1/3 of those in mother.	It diffuses readily into all body fluids (cerebrospinal, pleural and ascitic fluids), tissues, organs and excreta (saliva, sputum and faeces). It crosses the placenta and is secreted in the milk.
Metabolism		
General	Primarily hepatic, rapidly deacetylated.	Hepatic; primarily acetylated by – acetyltransferase to –N acetylisoniazid
Active metabolite(s)	25-o-deacetyl rifampicin	Nicotinoyl-NAD adduct
Elimination		
Elimination half life	3-5 hours Decreases to 2-3 hours after repeated administration	1.2 hours: rapid acetylators 3.5 hours: Slow acetylators
Mean systemic clearance (Cl/F)	5.7- 9 .0 L/hour	15.5 L/hour: slow NAT2 genotype 26.1 L/hour: rapid/ intermediate NAT2 genotype
% of dose excreted in urine	30%	75-95%
% of dose excreted in faeces	60-65%	<10%
Pharmacokinetic linearity	Non linear	NA*
Drug interactions *(in vitro)*	Rifampicin induces hepatic enzymes	Isoniazid is CYP450 inducer and inhibitor. Isoniazid is a arylamine n-acetyltransferase 2 substrate and inhibitor
Transporters	Solute carrier transporters (SLC) ATP Binding Cassette transporters (ABC) P-glycoprotein 1	NA*
Metabolizing enzymes	CYP450	CYP450: 2C19, 3A4

*NA information not available

Special populations

Rifampicin

The half-life of rifampicin has been reported to be longer in patients with liver impairment or biliary obstruction.

The half-life does not differ in patients with renal failure at doses not exceeding 600 mg daily, and consequently, no dosage adjustment is required. The half-life of rifampin at a dose of 720 mg daily has not been established in patients with renal failure. Following a single 900 mg oral dose of rifampin in patients with varying degrees of renal insufficiency, the mean half-life increased from 3.6 hours in healthy adults to 5.0, 7.3, and 11.0 hours in patients with glomerular filtration rates of 30 to 50 mL/min, less than 30 mL/min, and in anuric patients, respectively.

In one study, pediatric patients 6 to 58 months old were given rifampin suspended in simple syrup or as dry powder mixed with applesauce at a dose of 10 mg/kg body weight. Peak serum concentrations of 10.7 ± 3.7 and 11.5 ± 5.1 mcg/mL were obtained 1 hour after preprandial ingestion of the drug suspension and the applesauce mixture, respectively. After the administration of either preparation, the t1/2 of rifampin averaged 2.9 hours. It should be noted that in other studies in pediatric populations, at doses of 10 mg/kg body weight, mean peak serum concentrations of 3.5 mcg/mL to 15 mcg/mL have been reported.

Isoniazid

In slow acetylators with severely impaired renal function, accumulation of isoniazid may occur.
An impaired liver function prolongs the elimination half-life of isoniazid.

INDICATIONS AND USAGE

For pulmonary tuberculosis in which organisms are susceptible, and when the patient has been titrated on the individual components and it has therefore been established that this fixed dosage is therapeutically effective.
This fixed-dosage combination drug is not recommended for initial therapy of tuberculosis or for preventive therapy.
In the treatment of tuberculosis, small numbers of resistant cells, present within large populations of susceptible cells, can rapidly become the predominating type. Since rapid emergence of resistance can occur, culture and susceptibility tests should be performed in the event of persistent positive cultures.
This drug is not indicated for the treatment of meningococcal infections or asymptomatic carriers of *N. meningitidis* to eliminate meningococci from the nasopharynx.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

Akurit is contraindicated in patients with previous hypersensitivity reaction to any of the rifamycins or to isoniazid, including drug-induced hepatitis. It is contraindication in patients with previous isoniazid-associated hepatic injury; severe adverse reactions to isoniazid, such as fever, chills and arthritis; acute liver disease of any etiology.

WARNINGS

Akurit (Rifampin and Isoniazid Tablets) is a combination of two drugs, each of which has been associated with liver dysfunction. Liver function tests should be performed prior to therapy with Akurit and periodically during treatment.

Rifampin

Rifampin has been shown to produce liver dysfunction. There have been fatalities associated with jaundice in patients with liver disease or receiving rifampin concomitantly with other hepatoxic agents. Since an increased risk may exist for individuals with liver disease, benefits must be weighed carefully against the risk of further liver damage.
Several studies of tumorigenicity potential have been done in rodents. In one strain of mice known to be particularly susceptible to the spontaneous development of hepatomas, rifampin given at a level 2-10 times the maximum dosage used clinically resulted in a significant increase in the occurrence of hepatomas in female mice of this strain after one year of administration.
There was no evidence of tumorigenicity in the males of this strain, in males or females of another mouse strain, or in rats.

Isoniazid

Severe and sometimes fatal hepatitis associated with isoniazid therapy may occur and may develop even after many months of treatment. The risk of developing hepatitis is age related. Therefore, patients given isoniazid should be carefully monitored and interviewed at monthly intervals.

PRECAUTIONS

Rifampin

Rifampin is not recommended for intermittent therapy; the patient should be cautioned against intentional or accidental interruption of the daily dosage regimen since rare renal hypersensitivity reactions have been reported when therapy was resumed in such cases. Rifampin has been observed to increase the requirements for anticoagulant drugs of the coumarin type. The cause of the phenomenon is unknown. In patients receiving anticoagulants and rifampin concurrently, it is recommended that the prothrombin time be performed daily or as frequently as necessary to establish and maintain the required dose of anticoagulant.
Urine, feces, saliva, sputum, sweat, and tears may be colored red-orange by rifampin and its metabolites. Soft contact lenses may be permanently stained. Individuals to be treated should be made aware of these possibilities.
It has been reported that the reliability of oral contraceptives may be affected in some patients being treated for tuberculosis with rifampin in combination with at least one other antituberculosis drug. In such cases, alternative contraceptive measures may need to be considered.
It has also been reported that rifampin given in combination with other antituberculosis drugs may decrease the pharmacologic activity of methadone, oral hypoglycemics, digitoxin, quinidine, disopyramide, dapsone, and corticosteroids. In these cases, dosage adjustment of the interacting drugs is recommended.
Therapeutic levels of rifampin have been shown to inhibit standard microbiological assays for serum folate and vitamin B_{12} . Alternative methods must be considered when determining folate and vitamin B_{12} concentrations in the presence of rifampin. Since rifampin has been reported to cross the placental barrier and appear in cord blood and in maternal milk, neonates and newborns of rifampin-treated mothers should be carefully observed for any evidence of untoward effects.

Isoniazid

All drugs should be stopped and an evaluation of the patient should be made at the first sign of a hypersensitivity reaction.
Use of isoniazid should be carefully monitored in the following:

1. Patients who are receiving phenytoin (diphenylhydantoin) concurrently. Isoniazid may decreasethe excretion of phenytoin or may enhance its effects. To avoid phenytoin intoxication, appropriate adjustment of the anticonvulsant dose should be made.
2. Daily users of alcohol. Daily ingestion of alcohol may be associated with a higher incidence of isoniazid hepatitis.
3. Patients with current chronic liver disease or severe renal dysfunction.

Periodic ophthalmoscopic examination during isoniazid therapy is recommended when visual symptoms occur.
Interactions with other medicinal products and other forms of interaction:
Akurit is contraindicated in patients with previous hypersensitivity reaction to any of the rifamycins or to isoniazid, including drug-induced hepatitis. It is contraindication in patients with previous isoniazid-associated hepatic injury; severe adverse reactions to isoniazid, such as fever, chills and arthritis; acute liver disease of any etiology.

INTERACTION WITH OTHER MEDICAMENTS

Rifampicin is a very potent inducer of the hepatic and intestinal cytochrome P-450 enzyme system, as well as of glucuronidation and the P-glycoprotein transport system. Administration of rifampicin with drugs that undergo biotransformation through these metabolic pathways is likely to accelerate elimination of coadministered drugs. These effects approach their maximum after about 10 days of treatment, and gradually return to normal within 2 or more weeks after discontinuation. This must be taken into account when cotreating with other drugs. To maintain optimum therapeutic blood levels, dosages of drugs metabolized by these enzymes may require adjustment when starting or stopping the concomitant administration of Akurit.
In vitro, isoniazid acts as an inhibitor of CYP2C19 and CYP3A4. Thus it may increase exposure to drugs mainly eliminated through either of these pathways. However, when co-treating with rifampicin, as when using Akurit, these effects are likely to be outweighed by the hepatic enzyme induction due to rifampicin. Insofar as it has been investigated, the net effect of rifampicin and isoniazid on drug clearance will be an increase due to rifampicin rather than a decrease due to isoniazid.
Concurrent use of isoniazid with other hepatotoxic or neurotoxic medications may increase the hepatotoxicity and neurotoxicity of isoniazid, and should be avoided.
Mainly due to rifampicin, Akurit may interact with a very large number of other drugs, primarily by reducing the exposure to coadministered agents, reducing their efficacy and increasing the risk of therapeutic failure. For many important therapeutic agents, no interaction data with rifampicin are available. However, clinically significant reductions in drug exposure may occur. Whenever co-prescribing any drug together with Akurit the possibility of a drug-drug interaction should be considered. The following list of drug interactions with Akurit is not exhaustive, but is a selection of interactions of putative importance. The scope of the table is largely based on the WHO Essential Medicines List.
Use of isoniazid should be carefully monitored with patients with current chronic liver disease. Severe and sometimes fatal hepatitis associated with isoniazid therapy may occur and may develop even after many months of treatment.

Drugs by Therapeutic Area	Interaction	Recommendations concerning co-administration
INFECTION *Antiretrovirals*		
Nucleoside analogues **Zidovudine** / rifampicin	Zidovudine AUC ↓ 47%	The clinical significance of the owered zidovudine exposure is unknown. Dose modifications of zidovudine in this situation have not been formally evaluated.
Stavudine Didanosine Lamivudine Emtricitabine	No interaction expected	No dose adjustment required.
Tenofovir alafenamide/ emtricitabine/ rifampicin	Interaction not studied. Co-administration of rifampicin, a P-gp inducer, may decrease tenofovir alafenamide plasma concentrations, which may result in loss of therapeutic effect and development of resistance.	Coadministration is not recommended.
Tenofovir disoproxil / rifampicinTenofovir	AUC ↓13%	No dose adjustment required.
Abacavir / rifampicin	Empirical data are lacking, but rifampicin may decrease abacavir exposure through induction of glucuronidation.	Efficacy of abacavir should be closely monitored in co-treatment.
Non-nucleoside analogues		
Efavirenz / rifampicin	Efavirenz AUC ↓ 26%	When co-treating with AKuriT, it may be considered to increase the efavirenz dose to 800 mg q.d.
Nevirapine / rifampicin	nevirapine: AUC ↓ 58%	Since neither appropriate doses of nevirapine, when given concomitantly with rifampicin, nor the safety of this combination have been established, concomitant use of AKuriT and nevirapine is not recommended.
Etravirine / rifampicin	Rifampicin is likely to significantly reduce exposure to etravirine.	Co-treatment of AKuriT and etravirine should be avoided.
Protease inhibitors **Fosamprenavir** / rifampicin **Saquinavir Indinavir Ritonavir Lopinavir Atazanavir Tipranavir Darunavir Boceprevir**	Protease inhibitor exposure will be reduced to subtherapeutic level due to interaction with rifampicin. Attempts to dose adjust by increased doses, or an increase in ritonavir-boosting, have been ineffective or ill-tolerated with a high rate of hepatotoxicity.	AKuriT must not be co-administered with HIV or HCV protease inhibitors (see section 4.3).
Others		
Raltegravir / rifampicin	Raltegravir AUC ↓ 40%	Co-treatment should be avoided. If deemed necessary, consider an increase of the raltegravir dose to 600 mg b.i.d.
Dolutegravir / rifampicin	Dolutegravir AUC ↓ 54%	A dose adjustment of dolutegravir to 50 mg twice daily is recommended when coadministered with AKuriT in the absence of integrase class resistance. In the presence of integrase class resistance this combination should be avoided.
Elvitegravir/cobicistat/ rifampicin	Coadministration has not been studied. Rifampicin is a potent inducer of CYP450 metabolism and may cause significant decrease in the plasma concentration of elvitegravir and cobicistat resulting in loss of therapeutic effect.	Coadministration is contraindicated.
Maraviroc / rifampicin	Maraviroc AUC ↓ 63%	Co-treatment should be avoided. If deemed necessary, the maraviroc dose should be increased to 600 mg b.i.d.
Antivirals Hepatitis C-infection		
Daclatasvir Elbasvir/Grazoprevir Glecaprevir/Pibrentasvir Ledipasvir/Sofosbuvir Ombitasvir/paritaprevir/ ritonavir (with or without dasabuvir) Simeprevir Sofosbuvir (with or without velpatasvir with or without voxilaprevir)/ Rifampicin Isoniazid	Rifampicin: Coadministration has not been studied but is expected to decrease concentrations of these HCV-antivirals due to induction of CYP3A4 by rifampicin and hence to reduce their therapeutic effect. Isoniazid: Coadministration has not been studied. Patients with current chronic liver disease should be carefully monitored. Severe and sometimes fatal hepatitis associated with isoniazid therapy may develop even after many months of treatment.	Coadministration of AKuriT with these antivirals is contraindicated (for further details see Summary of product characteristics of the drugs for therapy of HCV).
Antifungals		
Ketoconazole / rifampicin	Ketoconazole AUC ↓ 80%	Co-administration should be avoided. If deemed necessary, a dose increase of ketoconazole may be required.
Fluconazole / rifampicin	Fluconazole AUC ↓ 23%	Monitor therapeutic effect.. An increased dose of fluconazole may be required.
Itraconazole / rifampicin	Itraconazole AUC ↓ >64-88%	Co-administration should be avoided.
Voriconazole / rifampicin	Voriconazole AUC ↓ 96%	Co-administration is contraindicated. If necessary, rifabutin should be substituted for rifampicin.
Antibacterials/Antituberculotics		
Clarithromycin / rifampicin	Clarithromycin mean serum concentration ↓ 85%. 14-OH clarithromycin levels	Co-administration should be unchanged. avoided.
Chloramphenicol / rifampicin	Case reports indicate >60-80% reduction of chloramphenicol exposure.	Co-administration should be avoided.
Ciprofloxacin / rifampicin	No significant interaction	No dose adjustment required.
Doxycyclin / rifampicin	Doxycyclin AUC ↓ 50- 60%	If co-treatment is considered necessary, the dose of doxycyclin should be doubled.
Metronidazole / rifampicin	Metronidazole AUC i.v. ↓ 33%	The clinical relevance of the interaction is unknown. No dose adjustment is routinely recommended. Efficacy should be monitored.
Sulfamethoxazole / rifampicin	Sulfamethoxazole AUC ↓ 23%	Interaction probably not clinically significant. Efficacy of sulfamethoxazole should be monitored.
Trimethoprim / rifampicin	Trimethoprim AUC ↓ 47%	A dose increase of trimethoprim may be required. Efficacy should be monitored.
Ethionamide / rifampicin		Rifampicin and ethionamide should not be co-administered, due to an increased risk of hepatotoxicity.
Antimalarials		
Chloroquine / rifampicin		Empirical data are not available. Since chloroquine undergoes polymorphic hepatic metabolism, lower levels are likely during rifampicin co- therapy. Co-administration should be avoided.
Atovaquone / rifampicin	Atovaquone AUC ↓ 50% Rifampicin AUC ↓ 30%	Co-administration should be avoided.
Mefloquine / rifampicin	Mefloquine AUC ↓ 68%	Co-administration should be avoided.
Amodiaquine / rifampicin	Empirical data are not available. Since amodiaquine undergoes hepatic metabolism, it is likely that clearance is increased when co-treating with rifampicin.	Co-administration should be avoided.
Quinine / rifampicin	AUC ↓ ≈ 80%. This has been associated with significantly higher recrudescence rates.	Co-administration should be avoided. If co-administration is deemed necessary, an increased dose of quinine should be considered.
Lumefantrine / rifampicin	Lumefantrine AUC ↓ 68%	Co-administration should be avoided.
Artemisinin and its derivatives / rifampicin	Artemether AUC ↓ 89% Dihydroarthemisinin AUC ↓ 85%	Co-administration should be avoided.
ANALGESICS, ANTIPYRETICS, NON- STEROIDAL ANTI-INFLAMMATORY DRUGS		
`**Morphine** / rifampicin	Morphine AUC p.o ↓ 30%, loss of analgesic effect.	Co-treatment should be avoided. If deemed necessary, efficacy should be monitored and the dose may need to be increased.
Codeine / rifampicin	Plasma levels of morphine, the active moiety of codeine, are likely to be substantially reduced.	Efficacy should be monitored and codeine dose increased if necessary.
Methadone / rifampicin	Methadone AUC ↓ 33-66%	Patients should be monitored for possible withdrawal effects, and methadone dose increased as appropriate (up to 2-3 fold).
Acetaminophen (paracetamol)/ rifampicin / isoniazid	Rifampicin may increase the glucuronidation of paracetamol and decrease the efficacy. There may be an increased risk of hepatotoxicity on co-administration, but data are inconclusive. Concurrent use with isoniazid may increase hepatotoxicity.	Co-administration of AKuriT and acetaminophen (paracetamol) should be avoided.
ANTICONVULSANTS		
Carbamazepine / rifampicin/ isoniazid	Rifampicin is expected to decrease the serum concentration of carbamazepine whereas isoniazid may increase them. Neurological side effects and the. risk of hepatotoxicity increase when co-treating with carbamazepine.	Co-administration of AKuriT and carbamazepine should be avoided.
Phenobarbital / rifampicin / isoniazid	Phenobarbital and rifampicin are both strong hepatic enzyme inducers, and each drug may lower the plasma concentrations of the other. Also, co- treatment with phenobarbital and isonazid may increase the risk of hepatotoxicity.	Co-administration of AKuriT and phenobarbital should be undertaken with caution, including monitoring of clinical effects and, if possible, plasma drug concentrations.
Phenytoin / rifampicin	Phenytoin AUC i.v. ↓ 42% Co-treatment with isoniazid phenytoin and isoniazid may result in an increased risk of hepatotoxicity.	Co-treatment with phenytoin and AKuriT should be avoided.
Valproic acid / rifampicin	Interaction studies are lacking. Since valproic acid is eliminated through hepatic metabolism, including glucuronidation, reduced plasma levels of valproic acid are likely with concomitant use.	Co-treatment should be avoided. If deemed necessary, efficacy and, if possible, also plasma concentrations of valproic acid, should be carefully monitored.
Lamotrigine / rifampicin	Lamotrigine AUC ↓ 45%	Co-treatment should be avoided. If deemed necessary, lamotrigine dose should be increased as appropriate.
IMMUNOSUPPRESSIVES		
Cyclosporine / rifampicin	Several studies and case reports have shown substantially increased cyclosporine clearance when co-administered with rifampicin.	Co-administration should be avoided. If deemed necessary, plasma concentrations of cyclosporine should be monitored and doses adapted accordingly (3-5 fold increases in cyclosporine dose have been required).
Tacrolimus / rifampicin **Sirolimus Everolimus**	Tacrolimus AUC i.v. ↓ 35%; AUC p.o ↓ 6870% Sirolimus AUC ↓ 82% Everolimus AUC ↓ 63%	Co-administration of AKuriT and mTOR inhibitors should be avoided. If deemed necessary, plasma drug concentrations should be monitored, and the dose increased as appropriate.
CARDIOVASCULAR MEDICINES		
Warfarin / rifampicin /isoniazid	Warfarin AUC ↓ 85% Isoniazid may inhibit hepatic metabolism of warfarin.	Monitor closely and adjust warfarin dose as needed and reduce dose after withdrawing rifampicin treatment.
Atenolol / rifampicin	Atenolol AUC ↓ 19%	No dose adjustment required.
Verapamil / rifampicin	S-verapamil p.o CL/F ↓ 32- fold. With i.v. S-verapamil, CL ↓ 1.3-fold	AKuriT and verapamil per-orally should not be co-administered. If i.v. verapamil is given, the therapeutic effect should be carefully monitored; dose adjustment may be required.
Digoxin / rifampicin	AUC p.o ↓ 30%	When co-administering AKuriT with digoxin, the efficacy and plasma concentration of digoxin should be monitored. A dose increase may be required.
Lidocaine. / rifampicin	Lidocaine CLi.v. ↓ 15%	No dose adjustment required
Amlodipine / rifampicin	Amlodipine, like other calcium channel blockers, is metabolised by CYP3A; lower exposure is expected when co-treating with rifampicin.	Efficacy should be monitored.
Enalapril / rifampicin	No interaction expected	No dose adjustment required.
Simvastatin / rifampicin	Simvastatin AUC ↓ 87% Simvastatin acid AUC ↓ 93%	Co-administration is not recommended.
Atorvastatin / rifampicin	Atorvastatin AUC ↓ 80%	Co-administration is not recommended
GASTROINTESTINAL MEDICINES		
Ranitidine / rifampicin	Ranitidine AUC ↓ 52%	Efficacy should be monitored, and ranitidine dose increased if necessary.
Antacids / isoniazid / rifampicin	Antacids may reduce the bioavailability of rifampicin by up to one third. Aluminium hydroxide impairs the absorption of isoniazid.	The clinical importance is unknown. Acid-suppressing drugs or antacids that do not contain aluminium hydroxide should be used, if co-treatment with AKuriT is necessary.
PSYCHOTHERAPEUTIC MEDICINES		
Diazepam / rifampicin / isoniazid **Midazolam Triazolam Alprazolam Nitrazepam**	Diazepam AUC ↓ >70% Midazolam AUC ↓ 98% Triazolam AUC ↓ 95% Alprazolam AUC ↓ 88% Reduced nitrazepam through concentrations, increased clearance.	Co-treatment is not recommended. Benzodiazepine withdrawal may occur in dependent individuals.
Zolpidem / rifampicin **Zopiclone** /rifampicin	Zolpidem AUC ↓ 73% Zopiclone AUC ↓ 82%	Co-administration should be avoided.
Chlorpromazine /rifampicin / isoniazid	Rifampicin may reduce chlorpromazine exposure. Also, concomitant use of chlorpromazine with isoniazid may impair the metabolism of isoniazid.	Co-administration should be avoided. If considered necessary, patients should be carefully monitored for isoniazid toxicity.
Haloperidol / rifampicin **Clozapine**	Haloperidol clearance is substantially increased by rifampicin, theoretical considerations imply that same applies to clozapine.	If co- treatment of AKuriT with haloperidol or clozapine is deemed necessary, monitor clinical efficacy. A dose increase may be required.
Amitriptyline / rifampicin **Nortriptyline**	Case reports (supported by theoretical considerations) suggest that rifampicin considerably increases clearance of tricyclic antidepressants.	Co-treatment should be avoided. If necessary, monitor for clinical response, side effects, and, if possible, plasma concentrations.
HORMONES; OTHER ENDOCRINE MEDICINES AND CONTRACEPTIVES		
Prednisolone / rifampicin **And other systemically administered corticosteroids**	Prednisolone AUC ↓ 66% Also for other corticosteroids, exposure is likely to be substantially decreased when co-treating with rifampicin.	Co-administration of AKuriT with corticosteroids should be avoided. If deemed necessary, the clinical status of the patient should be carefully monitored, and corticosteroid doses adjusted as needed.
Glibenclamide / rifampicin **Glimepiride Repaglinide**	Glibenclamide AUC ↓ 39% Glimepiride AUC ↓ 34% Repaglinide AUC ↓ 57%	Blood glucose levels should be closely monitored. A dose increase of diabetes medication may be required.
Insulin	No interaction expected.	No dose adjustment required.
Levothyroxine / rifampicin	Case reports indicate that rifampicin may decrease the effect of levothyroxine. .	TSH levels should be monitored
Ethinylestradiol / rifampicin	Ethinylestradiol AUC ↓ 66%	Co-adminstration with AKuriT may be associated with decreased contraceptive efficacy. Barrier- or other non-hormonal methods of contraception should used.
Norethindrone / rifampicin	Norethindrone AUC ↓ 51%	Co-administration with AKuriT may be associated with decreased contraceptive efficacy. Barrier- or other non- hormonal methods of contraception should be used.
OTHERS		
Praziquantel / rifampicin	Praziquantel AUC ↓ 80- 99%	Co-treatment with AKuriT should be avoided.
Disulfiram / isoniazid	Concurrent use of disulfiram together with isoniazid may result in increased incidence of adverse effects on the central nervous system.	Dose reduction or discontinuation of disulfiram may be necessary during therapy with AKuriT.
Theophylline / Isoniazid / Rifampicin	Isoniazid may increase the serum concentration of theophylline and rifampicin may increase it. The effects of combination are unknown.	Theophylline dose adjustment may be needed.
Enflurane / Isoniazid	Isoniazid may increase the formation of the potentially nephrotoxic inorganic fluoride metabolite of enflurane.	Coadministration of AKuriT with enflurane should be avoided.

PREGNANCY AND LACTATION

Rifampin

Although rifampin has been reported to cross the placental barrier and appear in cord blood, the effect of rifampin, alone or in combination with other antituberculosis drugs, on the human fetus is not known. An increase in congenital malformations, primarily spina bifida and cleft palate, has been reported in the offspring of rodents given oral doses of 150-250 mg/kg/day of rifampin during pregnancy.
The possible teratogenic potential in women capable of bearing children should be carefully weighed against the benefits of therapy.

Isoniazid

It has been reported that in both rats and rabbits, isoniazid may exert an embryocidal effect when administered orally during pregnancy, although no isoniazid-related congenital anomalies have been found in reproduction studies in mammalian species (mice, rats, and rabbits). Isoniazid should be prescribed during pregnancy only when therapeutically necessary. The benefit of preventive therapy should be weighed against a possible risk to the fetus. Preventive treatment generally should be started after delivery because of the increased risk of tuberculosis for new mothers.
Since isoniazid is known to cross the placental barrier and to pass into maternal breast milk, neonates and breast-fed infants of isoniazid treated mothers should be carefully observed for any evidence of adverse effects.
Carcinogenesis : Isoniazid has been reported to induce pulmonary tumors in a number of strains of mice.

ADVERSE EFFECTS

Rifampin

Nervous system reactions: headache, drowsiness, fatigue, ataxia, dizziness, inability to concentrate, mental confusion, visual disturbances, muscular weakness, pain in extremities, and generalized numbness.
Gastrointestinal disturbances: in some patients heartburn, epigastric distress, anorexia, nausea, vomiting, gas, cramps, and diarrhea.
Hepatic reactions: transient abnormalities in liver function tests (e.g., elevations in serum bilirubin, BSP, alkaline phosphatase, serum transaminases) have been observed. Rarely, hepatitis or a shocklike syndrome with hepatic involvement and abnormal liver function tests.
Renal reactions: elevations in BUN and serum uric acid have been reported. Rarely, hemolysis, hemoglobinuria, hematuria, interstitial nephritis, renal insufficiency, and acute renal failure have been noted. These are generally considered to be hypersensitivity reactions. They usually occur during intermittent therapy or when treatment is resumed following intentional or accidental interruption of a daily dosage regimen, and are reversible when rifampin is discontinued and appropriate therapy instituted.
Hematologic reactions: thrombocytopenia, transient leukopenia, hemolytic anemia, eosinophilia, and decreased hemoglobin have been observed. Thrombocytopenia has occurred when rifampin and ethambutol were administered concomitantly according to an intermittent dose schedule twice weekly and in high doses.
Allergic and immunological reactions: occasionally pruritus, urticaria, rash, pemphigoid reaction, eosinophilia, sore mouth, sore tongue, and exudative conjunctivitis. Rarely, hemolysis, hemoglobinuria, hematuria, renal insufficiency or acute renal failure have been reported which are generally considered to be hypersensitivity reactions. These have usually occurred during intermittent therapy or when treatment was resumed following intentional or accidental interruption of a daily dosage regimen and were reversible when rifampin was discontinued and appropriate therapy instituted. Although rifampin has been reported to have an immunosuppressive effect in some animal experiments, available human data indicate that this has no clinical significance.
Metabolic reactions: elevations in BUN and serum uric acid have occurred.
Miscellaneous reactions: fever and menstrual disturbances have been noted.

Isoniazid

The most frequent reactions are those affecting the nervous system and the liver.
Nervous system reactions: peripheral neuropathy is the most common toxic effect. It is dose-related, occurs most often in the malnourished and in those predisposed to neuritis (e.g., alcoholics and diabetics), and is usually preceded by paresthesias of the feet and hands. The incidence is higher in "slow inactivators." Other neurotoxic effects, which are uncommon with conventional doses, are convulsions, toxic encephalopathy, optic neuritis and atrophy, memory impairment, and toxic psychosis.
Gastrointestinal reactions: nausea, vomiting, epigastric distress and pancreatitis.
Hepatic reactions: elevated serum transaminases (SGOT; SGPT), bilirubinemia, bilirubinuria, jaundice, and occasionally severe and sometimes fatal hepatitis. The common prodromal symptoms are anorexia, nausea, vomiting, fatigue, malaise, and weakness. Mild and transient elevations of serum transaminase levels occurs in 10 to 20 percent of persons taking isoniazid. The abnormality usually occurs in the first 4 to 6 months of treatment but can occur at any time during therapy. In most instances, enzyme levels return to normal with no necessity to discontinue medication. In occasional instances, progressive liver damage occurs, with accompanying symptoms. In these cases, the drug should be discontinued immediately. The frequency of progressive liver damage increases with age. It is rare in persons under 20, but occurs in up to 2.3 percent of those over 50 years of age.
Hematologic reactions: agranulocytosis, hemolytic sideroblastic or aplastic anemia, thrombocytopenia, and eosinophilia
Hypersensitivity reactions: fever, skin eruptions (morbilliform, maculopapular, purpuric, or exfoliative), lymphadenopathy, and vasculitis
Metabolic and endocrine reactions: pyridoxine deficiency, pellagra, hyperglycemia, metabolic acidosis, and gynecomastia
Miscellaneous reactions: rheumatic syndrome and systemic lupus erythematosus-likesyndrome

OVERDOSE

Signs and Symptoms

Rifampin

Nausea, vomiting, and increasing lethargy will probably occur within a short time after ingestion; actual unconsciousness may occur with severe hepatic involvement. Brownish-red or orange discoloration of the skin, urine, sweat, saliva, tears, and feces is proportional to amount ingested.
Liver enlargement, possibly with tenderness, can develop within a few hours after severe overdosage, and jaundice may develop rapidly. Hepatic involvement may be more marked in patients with prior impairment of hepatic function. Other physical findings remain essentially normal.
Direct and total bilirubin levels may increase rapidly with severe overdosage; hepatic enzyme levels may be affected, especially with prior impairment of hepatic function. A direct effect upon hemopoietic system, electrolyte levels, or acid-base balance is unlikely.

Isoniazid

Isoniazid overdosage produces signs and symptoms within 30 minutes to 3 hours. Nausea, vomiting, dizziness, slurring of speech, blurring of vision, visual hallucinations (including bright colors and strange designs), are among the early manifestations. With marked overdosage, respiratory distress and CNS depression, progressing rapidly from stupor to profound coma, are to be expected, along with severe, intractable seizures. Severe metabolic acidosis, acetonuria, and hyperglycemia are typical laboratory findings.

Treatment

The airway should be secured and adequate respiratory exchange established. Only then should gastric emptying (lavage-aspiration) be attempted; this may be difficult because of seizures. Since nausea and vomiting are likely to be present, gastric lavage is probably preferable to induction of emesis.
Activated charcoal slurry instilled into the stomach following evacuation of gastric contents can help absorb any remaining drug in the GI tract. Antiemetic medication may be required to control severe nausea and vomiting.
Blood samples should be obtained for immediate determination of gases, electrolytes, BUN, glucose, etc. Blood should be typed and crossmatched in preparation for possible hemodialysis.
Rapid control of metabolic acidosis is fundamental to management. Intravenous sodium bicarbonate should be given at once and repeated as needed, adjusting subsequent dosage on the basis of laboratory findings (i.e., serum sodium, pH, etc.). At the same time, anticonvulsants should be given intravenously (i.e., barbiturates, diphenylhydantoin, diazepam) as required, and large doses of intravenous pyridoxine.
Forced osmotic diuresis must be started early and should be continued for some hours after clinical improvement to hasten renal clearance of drug and help prevent relapse. Fluid intake and output should be monitored.
Bile drainage may be indicated in presence of serious impairment of hepatic function lasting more than 24-48 hours. Under these circumstances and for severe cases, extracorporeal hemodialysis may be required; if this is not available, peritoneal dialysis can be used along with forced diuresis.
Along with measures based on initial and repeated determination of blood gases and other laboratory tests as needed, meticulous respiratory and other intensive care should be utilized to protect against hypoxia, hypotension, aspiration, pneumonitis, etc.
In patients with previously adequate hepatic function, reversal of liver enlargement and impaired hepatic excretory function probably will be noted within 72 hours, with rapid return toward normal thereafter.
Untreated or inadequately treated cases of gross isoniazid overdosage can terminate fatally, but good response has been reported in most patients brought under adequate treatment within the first few hours after drug ingestion.

Effects on ability to drive and use machines:

No studies on the effects on the ability to drive and use machines have been performed. Nevertheless, the clinical status of the patient and the adverse reaction profile of AKuriT, especially the potential neurotoxicity of isoniazid and nervous system reactions of Rifampicin should be borne in mind, which may impair patient's ability to drive or operate machinery.

DOSAGE AND ADMINISTRATION

In general, therapy should be continued until bacterial conversion and maximal improvement have occurred.
Adults: AkuriT is administered to patients on the basis of body weight.

Patient Body Weight (Kg)	CONTINUATION PHASE Four months **AkuriT tablets daily**
30-37	2
38-54	3
55-70	4
>70	5

Concomitant administration of pyridoxine (B_6) is recommended in the malnourished, in those predisposed to neuropathy (e.g., diabetic), and in adolescents.

PRESENTATION

Available as a pack Alu- Alu of 15 x 6 Tablets.
Also available as per user's request.

STORAGE: Store at temperature not exceeding 30°C.

Date of Revision: 20/05/2025

Manufactured by:
LUPIN LTD.
A-28/1, MIDC Area, Chikalthana,
Chhatrapati Sambhajinagar- 431210,
Maharashtra State, INDIA.